

ENGINEERING CHANGE REQUEST:

ECR - 3/1/2030:

Requestor: Prof. Kidd

Description of requested change: Request that the PROJECT-CE design be modified to completely replace the milli() function defined in arduino.h with one written by the design team. The requested change includes the accompanying register and other constants as well as the ISR. No change is requested to the Customer Requirements or the High-level requirements defined in the PROJECT-CE design document.

Reason for request: The proposed change is intended to

- (1) provide a more flexible design that the design team can customize and extend,
- (2) place the timing design entirely under company control and isolate it from future changes in third-party arduino.h, and
- (3) reduce possible future IP ownership claims. Although this may add initial development cost, the added independence, flexibility, extensibility and reuse are expected to reduce labor and costs in future projects.

DEFINITIONS:

"PROJECT-ASYNC", "PROJECT-CE" - The previous projects upon which this project is based

"arduino.h" - helper functions for the Arduino development platform maintained by Arduino

AG, owned by Qualcomm. IP - Intellectual Property

ISR - Interrupt Service Routine

PWM - Pulse Width Modulation

REQUESTED CHANGE SCOPE:

1. Replace the milli() functionality with a design-team implementation named mymilli().
2. Write a PWM counter ISR that provides the timing mechanism underlying mymilli().
3. Define the register constants needed by the timer implementation rather than relying on avr.h definitions.
4. Retain the PROJECT-CE system testing and validation requirements and add tests needed to demonstrate the modified system.
5. Revise the PROJECT-CE formal design document to identify the design changes and issue an ECN notifying affected teams.